Symmetrical Distribution

A symmetrical distribution is a probability distribution where the left side and right side of the graph are mirror images of each other around the center.

Mean = Median = Mode
Zero skewness
Equal spread on both sides of the center
Data values are evenly distributed around the central value

[2, 4, 6, 8, 10]

- Mean = 6
- Median = 6
- Values are evenly spaced around 6
- Left side (2,4) mirrors right side (8,10)

Distribution

A distribution describes how values of a variable are spread or arranged.

Probability Distribution

A probability distribution is a mathematical function that assigns a probability to each possible outcome of a random experiment.

1)**Discrete Probability Distribution:**
Used when outcomes are **countable**.
Rolling a die
Number of customers entering a store

2)Continuous Probability Distribution
Used when outcomes are **measurable** (infinite possible values).
Examples: Height, Weight, Time, Revenue.

Normal Distribution (Continuous)

The Normal distribution is a continuous probability distribution that is symmetric around its mean.

- Mean = Median = Mode
- Bell-shaped curve

Bernoulli Distribution (Discrete)

Bernoulli distribution is NOT continuous.
It is discrete and has only two outcomes.
Definition
The Bernoulli distribution models a single trial with:

- Success (1)
- Failure (0)

Binomial Distribution (Discrete)

The Binomial distribution models the number of successes in n independent Bernoulli trials.

Key Conditions

- Fixed number of trials (n)
- Two outcomes (success/failure)
- Constant probability (p)
- Independent trials

Uniform Distribution (Continuous)

- The **Uniform distribution** assumes:
- All values within a range have equal probability.

Poisson Distribution (Discrete)

- The **Poisson distribution** models the number of events occurring in a fixed interval of time or space.
- How many clicks per hour or minutes for a website

Bias:

In statistics and machine learning, bias does NOT mean "choosing the value that appears more."
Bias refers to systematic error — a consistent deviation from the true value.
If an estimator consistently overestimates or underestimates the true parameter, it is biased.

Descriptive Method (Descriptive Statistics)

The descriptive method is used to summarize, organize, and describe the main features of a dataset.
It does not make predictions.
It does not generalize to a population.
It only explains what the data shows.

Pearson's Correlation Coefficient (r)

Pearson's correlation coefficient measures the strength and direction of the linear relationship between two continuous variables.
Measures linear relationship only.

Correlation with Heatmap

A **correlation heatmap** is a visual representation of a **correlation matrix**, where color intensity represents the strength and direction of relationships between variables.
Detect multicollinearity
Understand variable relationships
Improve model interpretability

Cross-Validation

Cross-validation is a resampling technique used to evaluate how well a machine learning model generalizes to unseen or new data.

Hyperparameters are configuration settings of a model that are set **before training** and control how the model learns.
They are **not learned from data**.

Linear Regression
Logistic Regression
Decision Tree
Random Forest

Grid Search
Grid Search is a systematic method for finding the best hyperparameter combination by trying all possible combinations from a predefined grid.

To systematically tune hyperparameters
Combined with cross-validation to avoid overfitting
Ensures best model configuration within defined search space